GROUP 16 · VISIONARY

Signal.

One dashboard, every channel, one clear read.

A campaign-planning and measurement web app for marketing leads at early-stage startups, built as a coded clickable prototype for the Final Design Project.

LIVE PROTOTYPE

signal-drab-xi.vercel.app

COURSE

Human–Computer Interaction

INSTRUCTOR

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SUBMISSION

27 April 2026

GROUP

16 · Visionary

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REPOSITORY

github.com/Kazemkhani/HCI-Final-Project

Authorship & AI Use

In line with the course's AI Use Policy (Final Project Assignment, page 11), the team used generative AI tools to assist with drafting written sections of this report. The user research, persona, journey map, problem statements, How-Might-We questions (Stage 1), the prototype itself (Stage 3), the heuristic evaluation (Stage 4 Part B), the iteration fixes (Stage 5), and every individual reflection (Stage 6) are the team's own work — reviewed, edited, and owned by the members named on the cover page. Where AI assisted with phrasing or structuring, every line was read, revised, and signed off by a named team member before inclusion.

The Stage 1 evidence base was conducted by the team in February 2026 under informed consent, originally documented in Assignment 1, and is reproduced and extended here per the rubric's explicit allowance for continuing from Assignment 1. The Stage 4 usability sessions were conducted by the team between 21–23 April 2026 with five participants outside the group; participant codes T1–T5 are anonymised per our consent process. The repository at github.com/Kazemkhani/HCI-Final-Project carries the full commit trail, including `INTERACTIVITY_AUDIT.md` and `INTERACTIVITY.md` — the working documents that drove Stage 5 — for any claim that the marker wishes to verify.

Team Manifesto

Roles rotated across the project to ensure every member touched research, design, evaluation, and reflection.

Member	Primary Role	Contributions
Aquib Palampalliyali	Research Lead — Stage 1 + Stage 4 Part A	Re-cut interview evidence into the Stage 1 traceability matrix, recruited and ran usability testing sessions, scored SUS responses, drafted Sections 4.1 and 4.2.
Aman Riyas	Ideation Lead — Stage 2 + Stage 5	Ran the Crazy 8s workshop, facilitated dot-voting, drafted the competitive analysis section, owns the Stage 5 before/after iteration write-up.
Heng Cheng	Visual Design Lead — Stage 3	Annotated the prototype screen captures, rendered the user-flow diagram, owns the consolidation of the live `/style-guide` route into the report's Style Guide section.
Alagappan Alagappan	Evaluation Lead — Stage 4 Part B + QA	Led the heuristic evaluation of Signal mapped to Nielsen's 10 heuristics, compiled the consolidated issues table, ran rubric coverage QA on the final PDF.
Amir Hossein Kazemkhani	Prototype & Submission Lead	Built the coded front-end prototype (Next.js, deployed to Vercel), led the interactivity audit and bug-fix cycle that drove Stage 5 evidence, owns final formatting and Canvas upload.

Decision-making. Disagreements were resolved by structured discussion first; if no consensus emerged within ten minutes, we used dot voting (3 votes per member) on impact × feasibility — the same protocol that worked well in Assignment 2.

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Introduction

1.1 Project Scope

Signal is a campaign planning and measurement web app for marketing professionals at early-stage businesses. It addresses the persistent problem we identified in Assignment 1: marketers like our persona John Doe juggle multiple advertising channels — digital, physical, and hybrid — but lack a single, trustworthy view that tells them whether their spend is working. Our Final Design Project takes the validated problem space from Assignment 1 forward into a complete design solution: ideated against competitive alternatives, prototyped as a working coded front-end, evaluated through both heuristic review and usability testing with real participants, and iterated on the basis of evaluation findings.

The prototype lives at **signal-drab-xi.vercel.app**. The repository is open at github.com/Kazemkhani/HCI-Final-Project. This report is the rigorous account of how every design decision was made and what evidence drove each one.

1.2 Continuity from Assignment 1

Per the Final Project rubric, we continued the direction set in Assignment 1 rather than starting fresh. Assignment 1 produced a validated persona (John Doe, marketing lead at a Dubai startup) and five problem statements, four of which Signal directly addresses. The mapping from A1 problem statement to Signal feature is shown below; this table is the load-bearing traceability claim for our entire project.

A1 Problem Statement	Signal feature addressing it
PS1 — Channel Selection Overwhelm *"no standard framework for matching channels to audiences"*	`/channels` page with proposed mix, audience-match score, and live rebalance
PS2 — Resource Constraints Limiting Promotional Mix *small teams default to a narrow set of digital channels*	`/brief` 4-step wizard that produces a £-budget mix across digital, physical, and hybrid in under three minutes
PS3 — Cutting Through Noise *"audiences are exposed to constant competing messages"*	`/copy` Clarity Check using the Anthropic API to score ad copy against a stated goal
PS4 — Measuring Effectiveness Across Channels *physical channels provide no equivalent of digital metrics*	`/dashboard` cross-channel read with a single status colour per channel and a multi-channel trend chart — **the hero feature of Signal**
PS5 — Messaging That Resonates *"force-fitting logos, calls to action, and excessive details"*	`/copy` route's verdict-and-suggestion pattern that explicitly rewards simpler messaging

We did not pursue PS5's narrower stakeholder-pressure angle as a separate surface in this iteration; the Clarity Check addresses the symptom (cluttered messaging) rather than the underlying social dynamic. This is the only material direction change from A1; we explain it in Stage 5.

Stage 01

Stage 1 — Research & Problem Definition

Understanding John, his channels, and what is broken.

> Section 1 ports Assignment 1 evidence forward verbatim where the underlying claim has not changed; new content (design goals, evolution-from-A1) is marked.

1.1 Research Method & Participants

We used semi-structured interviews — a fixed three-band funnel of broad-to-specific questions with flexibility to probe unexpected leads, well suited to discovery research [1]. Four professionals across role, seniority and proximity to the problem space were interviewed 17–20 February 2026. All gave informed consent; data was anonymised P1–P4. Two interviews ran online due to scheduling constraints, approved by the course instructor. Full Q1–Q10 instrument and transcripts in **Appendix A**.

Code	Role	Experience	Date	Duration
P1	Founder, personal finance	4 yrs	17 Feb	11 min
P2	Startup founder, tech-leaning	2 startups	18 Feb	13 min
P3	Management academic + comms	15+ yrs	18 Feb	30 min
P4	Senior creative / advertising	15+ yrs	20 Feb	40 min

1.2 Interview Findings

The synthesis below names the load-bearing evidence each participant contributed. Per-interview tables and verbatim quote sets in **Appendix A**.

#	Load-bearing evidence	Reflected in Signal as
P1	“Categorise-and-ignore” loop on outdoor — most ads mentally filed before being read	Clarity Check (`/copy`) — scores ad copy on the *one thing the audience gains*
P2	Signups-as-primary-metric: “It doesn’t matter if you use the application. It matters how many people visit”	Dashboard KPI tiles — Spend, Signups, Cost per Signup, Audience Match
P3	“It depends” stance — refused single-best-channel framing; insisted on context-dependent mix with human in the loop	`/channels` surfaces a *proposed* mix with rationale; John adjusts
P4	CMO cuts outdoor not because it doesn’t work but because *no equivalent of clickthrough rates exists*	Cross-channel `/dashboard` — digital-equivalent metric layer over physical channels

1.3 Affinity Themes

Silent-affinity protocol: each member wrote one sticky per observation; we clustered without speaking, then labelled. Full FigJam board in **Appendix B**. Four themes promoted from six clusters:

1. **The "Noise" Barrier and Attention Scarcity.** Audiences ignore most advertising; the attention window is shrinking on every channel. *(P1, P2, P3, P4)*
2. **Resource Constraints Dictate Channel Strategy.** Financial and staffing limits force early-stage businesses toward digital regardless of audience fit. *(P2, P3, P4)*
3. **The Measurement Gap and Decision Agility.** Tension between needing to validate ROI and being unable to measure non-digital channels drives reactive decisions and a structural shift to digital. *(P2, P3, P4)*
4. **Messaging Simplicity vs. Stakeholder Pressure.** All four converged on simplicity as the property of effective messaging; stakeholder pressure consistently undermines it. *(P1, P2, P3, P4)*

1.4 Persona — John Doe

Synthesised from patterns across all four interviews. Every detail traces to interview evidence.

Field	Description
Name	John Doe — 28-year-old marketing lead, Dubai Marina
Background	BBA, 4 yrs experience, mid-sized Dubai startup, team of two, juggles multiple campaigns with limited budget
Goals	Build brand awareness efficiently; find the right channels for the right audience at the right time; prove ROI to founders/investors
Pain points	Overwhelmed by noise; budget excludes expensive channels; struggles to attribute results across non-digital formats
Behaviours	Defaults to digital because measurable; sceptical of flashy ads; pulls underperforming campaigns fast
Tools	Google Analytics, social-media management software, Canva, MacBook + iPhone
Key quote	*"Time is actually money. If the timing and audience aren't right, nothing else matters."*

1.5 Journey Map

Synthesised from patterns across all four interviews. Five sequential stages; the full action / thought / emotion / pain / touchpoint table is in **Appendix B**.

The map identifies five intervention points. Signal addresses three on the prototype's primary surfaces:

- **Choosing Channels** → `/brief` and `/channels`
- **Crafting the Message** → `/copy` (Clarity Check)
- **Measuring & Adapting** → `/dashboard` and `/budget`

The remaining two (Understanding the Audience, Executing the Campaign) are out of scope for this iteration, explained in Stage 6's reflection.

1.6 Problem Statements

Five problem statements in the form `As [persona], I want [job] but [obstacle] because [root cause], resulting in [consequence]`. Each cites at least two pieces of interview evidence; full long-form statements in **Appendix B**.

#	Headline	Evidence
PS1	Channel selection overwhelm — too many channels, no standard framework, decision paralysis	P3, P2, P4
PS2	Resource constraints — high cost and specialist expertise gate diversification	P3, P2, P4
PS3	Cutting through noise — oversaturated landscape, audiences habitually skip ads	P3, P2, P4
PS4	Measurement gap — no unified data across physical and digital, attribution unclear	P3, P2, P4
PS5	Messaging simplicity — stakeholders pressure messaging into clutter	P2, P3, P4

1.7 How Might We

Thirteen HMW questions across cognitive, behavioural, emotional, speed/time and trust dimensions. The seven below are the ones Signal's prototype directly answers; the remaining six (deferred) are listed in **Appendix B**.

- **Reduce the cognitive effort of comparing channels for a specific audience** *(PS1)* → `/channels` proposed mix + audience-match score
- **Increase confidence that a channel will reach the intended audience before committing spend** *(PS1)* → `/channels` per-channel rationale + 14-day trend
- **Enable effective multi-channel promotion within startup budget and team constraints** *(PS2)* → `/brief` sub-3-minute flow
- **Help John diversify beyond familiar digital channels without specialist expertise** *(PS2)* → hybrid and physical channels in the proposed mix
- **Capture attention within the 3–5 seconds audiences give** *(PS3)* → `/copy` Clarity Check
- **Make campaign performance across physical and digital channels comparable without external measurement firms** *(PS4)* → core thesis of `/dashboard`
- **Help John make confident budget reallocation decisions when attribution data is incomplete** *(PS4)* → `/budget`'s evidence-led recommendation flow

1.8 Insights and Design Goals

Three load-bearing insights, each tied to a measurable design goal Signal is assessed against.

Insight 1 — The Measurement Gap is the Root Cause. Physical channels are not abandoned because they don't work but because they don't *report* in the same vocabulary as digital. We design for *cross-channel comparability* before any single channel. → **Design Goal 1.** Compare cost-per-signup, signups and audience match across digital, physical and hybrid channels on a single screen, no scrolling, under 10 seconds. *(Evaluated in Stage 4: Task 2.)*

Insight 2 — Decision Agility is Asymmetric. All four participants pulled underperforming channels immediately, but none described a clean process for *moving* spend. We design the *budget reallocation* surface as a first-class flow. → **Design**

Goal 2. Review and approve a recommended budget move, with evidence and risk caveats, without leaving a single page, under 90 seconds. *(Evaluated in Stage 4: Task 3.)*

Insight 3 — Simplicity is Universally Demanded but Procedurally Defeated. All four named simplicity as the property of effective messaging; one named stakeholder pressure as the structural reason it loses. We design a tool that *takes the side of simplicity*. → **Design Goal 3.** Receive a structured verdict (0–100 score + one-sentence diagnosis + tighter rewrite) on ad copy under 6 seconds. *(Evaluated in Stage 4: Task 4.)*

Design Challenge. *How might we help time-pressed marketing leads at early-stage startups make confident, evidence-led decisions across physical and digital channels — without specialist tooling or external measurement firms?*

Stage 02

Stage 2 — Competitive Analysis & Ideation

Where the gaps are. How we converged on Signal.

2.1 Competitive Landscape

We analysed three commercial products in or adjacent to Signal's problem space. Each was evaluated against the rubric's three lenses: **what works**, **what fails**, **opportunities for Signal**, with Nielsen heuristics and Gestalt principles cited where relevant [2], [3].

2.1.1 HubSpot Marketing Hub

All-in-one marketing automation, mid-market and growth-stage.

What works	What fails	Opportunity for Signal
Single-source-of-truth campaign reporting (the cross-channel read Signal aspires to). Strong information scent — every tile shows a delta with directional arrow and contextual framing (Nielsen H1). Editorial rigour in attribution: first-touch / last-touch / linear models exposed transparently.	Monstrous IA (H8) — first-time marketing lead at a 5-person startup faces a 14-item sidebar, six paid upgrades. No first-class home for outdoor/experiential channels — paid ads UTM tracking covers Google/LinkedIn/Facebook well; everything else is "Other" (PS4 replicated inside the tool that claims to fix it). Cross-channel reporting gated at £750/month "Marketing Hub Professional" — a hard wall for the persona.	Be opinionated where HubSpot is configurable. Treat physical channels as first-class. Single consolidated dashboard, not 14 sidebar items. Cross-channel compare available by default, not paid-tier.

2.1.2 Mixpanel

Product analytics, used heavily by SaaS startups for funnel analysis.

What works	What fails	Opportunity for Signal
Time-to-first-insight is short — sane chart on selection of any event (Signal mirrors this in `/brief`'s pre-filled budget slider). Segmentation done well — Breakdown menu slices any metric by any property in one click; related controls cluster (Gestalt common region [4]). Cohorts are first-class — saving and reusing supports the "decision agility" insight.	Built for a single-channel reality; structurally unable to compare a podcast sponsorship to a LinkedIn ad — data isn't in the same database. Marketers must export to spreadsheets to reconstruct cross-channel — the failure mode Signal exists to fix. Interface is for analysts, not marketing leads (H6) — query language exposes funnel/event semantics requiring training. No spend or budget reallocation surface.	Borrow the "graph on first selection" defaults and Breakdown segmentation, but stay in marketing-lead vocabulary (channels, signups, audience match) not product-analyst vocabulary. Close the act-on-insight loop with `/budget`.

2.1.3 Northbeam

Marketing measurement for direct-to-consumer brands, multi-touch attribution.

What works	What fails	Opportunity for Signal
Single hero metric per view — “Cost per New Customer” leads, not buried among nine secondary tiles. Honest about confidence — annotates attribution with confidence intervals; tells the user when data is sparse (addresses P3’s concern that marketers default to confident-looking numbers). Excellent empty states — clear “what to do first” prompts, never an empty grid.	Locked to e-commerce and ads — podcast sponsorships, in-person events, PR placements absent or shoehorned into “offline” (PS4 again, inside a tool whose positioning claims to fix it). Onboarding requires Shopify + Meta Ads account — the persona we design for cannot use Northbeam at all. Visual density high; context buried behind hover states (H1 partial violation).	Match the hero-metric clarity and confidence-honesty without the e-commerce-only assumption. Design explicitly for the £10k/month, mixed-channels, no-Shopify reality of smaller-stage clients.

2.1.4 Cross-Cutting Opportunities

Three positioning gaps:

1. **No tool treats physical channels as first-class.** Signal’s `/dashboard` table puts LinkedIn Ads, Podcast Sponsorship and SaaStr Europa on the same row template — a structural commitment, not a UI choice.
2. **No tool closes the analyse-to-act loop in one surface.** Mixpanel tells; HubSpot reports; Northbeam attributes. None propose a specific budget move with evidence and a single approve/decline. Signal’s `/budget` is the differentiator.
3. **No tool serves the £10k/month marketing lead.** All three are priced and configured for in-house teams of 5–50. Signal’s defaults assume a one-person operation.

2.2 Crazy 8s — Individual Sketches

Standard Crazy 8s protocol [5] — A4 folded into 8 panels, 60 seconds per panel, no refining. Run **22 March 2026**, re-run **15 April 2026** once the early prototype revealed which surfaces still needed shaping. Photographs of all ten sheets and full per-member panel descriptions in **Appendix C**.

Member	Best-panel concept	One-line description
Aquib Palampalliyali	Channel Health Stripes	Horizontal green/amber/red stripes per channel row, thickness proportional to weekly spend.
Aman Riyas	Single-Move Recommendation Card	Full-width card: “Move £1,800 from LinkedIn Ads to Podcast” + 3 evidence rows + Approve/Decline.
Heng Cheng	Brief-as-Conversation	4-step progressive form, one question at a time, produces the channel mix.
Alagappan Alagappan	Clarity Check Verdict	Two-pane: original ad copy left; 0–100 score plus tightened rewrite right.
Amir Hossein Kazemkhani	One-Read Dashboard with Status Bar	4 KPI tiles + cross-channel table + left-edge coloured status bar. Reads in 10 seconds.

2.3 Concept Selection

We dot-voted on the five “best panel” concepts (3 votes per member, 15 total):

Concept	Author	Votes
One-Read Dashboard with Status Bar	Amir	5
Single-Move Recommendation Card	Aman	4
Channel Health Stripes	Aquib	3
Clarity Check Verdict	Alagappan	2
Brief-as-Conversation	Heng	1

Rather than pick one, we noticed the top three could be combined: One-Read Dashboard provides the surface, Recommendation Card lives inside as the act-on-insight CTA, Channel Health Stripes are the row-level visual encoding. Clarity Check Verdict became a utility route (`/copy`) — distinct task, not dashboard-bound. Brief-as-Conversation became `/brief` because the dashboard needs *something* to populate before it can render a meaningful state.

The combined concept is **Signal**:

> Signal is one dashboard that gives John an honest read across every channel, alerts him when something needs his attention, and walks him through a single budget move at a time — with the evidence and the caveats — until he approves it. Plus a separate Clarity Check that scores a piece of ad copy in under six seconds.

Rationale for the combined concept: maps to four of the five A1 problem statements (PS1 channels, PS3 noise, PS4 measurement, PS5 messaging — PS2 implicit in the £10k default and no external tooling); buildable as a coded prototype in two weeks; testable in a 30-minute usability session, each surface a discrete task. Dot-vote and combination minutes from 15 April 2026 in **Appendix C**.

Stage 03

Stage 3 — Prototype Design

The screens, the flow, the system.

> **Live prototype:** https://signal-drab-xi.vercel.app **Repository:** https://github.com/Kazemkhani/HCI-Final-Project **Stack:** Next.js 16 (App Router), TypeScript strict, Tailwind v4, shadcn/ui, Framer Motion, Recharts, Anthropic SDK for `/copy`. Deployed on Vercel.

3.1 Design Principles

Five principles, derived from the Stage 1 insights and design goals, committed before any screen was drawn:

1. **The dashboard is the product.** Everything else exists to populate or act on it. *(Insight 1, Design Goal 1.)*
2. **Physical channels render identically to digital ones.** Every row template, KPI tile and chart accepts a podcast sponsorship the same way it accepts a LinkedIn ad. No "secondary channels" tier at the data-model level. *(PS4.)*
3. **The act-on-insight loop is one click away.** Each dashboard alert contains a CTA that lands the user on a pre-populated `/budget` recommendation. *(Insight 2, Design Goal 2.)*
4. **Evidence is presented before conclusions.** No "AI ✨ recommends" boxes. The `/budget` page shows three pieces of evidence above the recommended action. *(P3's "it depends".)*
5. **British English, persona-first copy.** UI copy is written for John specifically. Eight words beat sixteen. *(Clarity Check measures this as a failure mode.)*

3.2 Annotated Key Screens

The prototype has seven routes; the four load-bearing ones are documented below. The remainder (`/brief`, `/flow`, `/style-guide`) are described in **Appendix G** with screenshots in `submission/materials/screenshots/`.

Screen 1 — `/dashboard` (Hero) → PS4, Design Goal 1

Four KPI tiles (Spend, Signups, Cost per Signup, Audience Match), cross-channel comparison table, alerts panel, multi-channel trend chart.

- **Date-range tabs (top right).** SegmentedTabs control; the active tab drives every downstream widget on the page (Stage-5 fix; original was a false affordance).
- **KPI delta with directional colour.** Up/down arrow + green/red — Nielsen H1, the user knows whether the change is good news without reading the label.
- **Status bar on every row.** Left-edge stripe (green / amber / red), redundantly encoded as a screen-reader label per WCAG 1.4.1.
- **Alert with embedded CTA.** "LinkedIn Ads cost per signup is up 8%" → "Review reallocation →" deep-links to `/budget?source=…&dest=…&amount=…`, pre-populated. Closes Principle 3 in one click.
- **Trend chart channel-coloured lines** match the row status-bar colours (Gestalt similarity [4]); hover dims non-focused lines.

Screen 2 — `/channels` (Mix and Rebalance) → PS1, PS2, HMW1, HMW4

Five channel cards, each with thumbnail, status pill, allocation, audience-match radial, "Why this channel" rationale, and a slider that proportionally rebalances the others (Zustand store).

- **Audience-match radial.** Tone reflects the score (≥80 green, 60–79 amber, <60 red). Replaced an earlier thick black bar that visually collided with the slider — Stage 5 fix.

- **"Adjust monthly allocation" zone.** Slider in a bordered region with min/max labels — telling the user this is a different *kind* of control from the buttons elsewhere.
- **Live mix bar (sidebar).** Stacked horizontal bar of the current mix, animated by Framer Motion as sliders move (Nielsen H1).
- **Sticky CTA footer.** "Launch campaign" sticks to the viewport bottom, reachable without scrolling regardless of card count.

Screen 3 — `/budget` (Single Move) → PS4, Design Goal 2, HMW10

A focused page that proposes one budget move at a time. Top: source → destination flow with before/after monetary values. Middle: three pieces of evidence. Bottom: sticky Decline / Adjust / Approve.

- **Flow visualisation.** Source channel left, destination right, animated arrow with the move amount; each card shows `before → after` with the old value struck through.
- **"Adjust" toggle gates the slider.** Hidden by default; clicking "Adjust" reveals it. Stage-5 fix — the original toggle's label flipped without any panel ever opening (false affordance).
- **Evidence is above the action.** Three evidence rows above the sticky footer — the user must scroll past them before reaching Approve. Slows the decision in the right place.
- **Decline modal echoes the reason.** The textarea's content is echoed in the confirmation toast — fixing a Stage-4 finding where input was silently discarded.
- **Things to watch.** Risk-caveat disclosure ("Podcast inventory books two weeks out") folded by default so caveats don't compete with primary evidence.

Screen 4 — `/copy` (Clarity Check) → PS3, PS5, Design Goal 3, HMW6

Two-pane: input on the left, result on the right. The user picks a goal, pastes ad copy, clicks "Score it." A real call to the Anthropic API (Claude Sonnet 4.5) returns a 0–100 clarity score, a one-sentence verdict, and a tightened rewrite.

- **Live API integration.** `app/api/clarity/route.ts` calls `@anthropic-ai/sdk` with `claude-sonnet-4-5`. Real, not simulated.
- **Cancel during loading.** "Score it" transforms into Cancel while in flight, wired to an AbortController (Nielsen H3).
- **Mid-band score colour.** A score of 65 reads as warning amber, not green or red — Stage-4 heuristic-eval fix.
- **Try-again on error.** API errors render an inline error state with a "Try again" button that re-fires the same input. Form fields are not wiped (Nielsen H9).

3.3 User Flow

The prototype is optimised for the five-step path `/brief → /channels → /dashboard → (/budget | /channels/[id] | /copy) → /dashboard`. The full diagram is documented graphically on `/flow`.

Path length. Brief to approved budget move = five surfaces. Stage 4 measured average path-completion at ~3 minutes 50 seconds.

Clarity Check is intentionally orthogonal. Reachable from the sidebar at any point; does not require the brief or any other state. P2's "give me one line of solution" can be tested in isolation.

3.4 Style Guide

A complete in-product style guide ships as a live route at `/style-guide` — it renders the actual tokens the app uses, so it cannot drift out of sync with a static PDF. Three tabs: System (colour ramp, type scale, 8px spacing grid, radii, elevation, motion), Components (buttons, status pills, IconBox, Disclosure, SegmentedTabs, EmptyState, RadialScore), Data Viz (Signal 5 chart palette — id-mapped, not index-mapped, so a channel's colour is stable across sort orders).

Token summary:

Token	Value	Use
`--color-bg`	`#FAFAF7`	Warm off-white background
`--color-accent`	`#10B981`	CTAs, focus, positive deltas
`--color-warning`	`#D97706`	“Attention” status
`--color-negative`	`#DC2626`	“Critical” status
`--color-editorial`	`#6D28D9`	Serif display moments only
Display font	Instrument Serif italic	Hero h1 only
Body font	Geist Sans 400/500/600	Everything else
Numbers	Geist Mono `tabular-nums`	KPIs, table cells
Spacing grid	`4, 8, 12, 16, 24, 32, 48, 64, 96, 128`	Strict 8px

Forbidden patterns: gradient hero backgrounds; “AI ✨ ” badges (we use the Anthropic API but don’t advertise it); stock photography (lucide-react icons only).

Accessibility baseline (verified in Stage 4): WCAG 2.2 AA contrast on every text/background pair; 2px focus ring with 2px offset on every interactive element; `aria-label` on all icon-only buttons; full keyboard completion (Tab + Enter); `prefers-reduced-motion: reduce` honoured.

Stage 04

Stage 4 — Evaluation

Five participants, four tasks, ten heuristic findings.

Two-part evaluation. Part A is a small-N moderated usability test with a System Usability Scale (SUS) questionnaire at the end of each session. Part B is a structured heuristic evaluation of Signal against Nielsen's 10 heuristics, applied using the same Phase 1 / Phase 2 methodology that worked well in Assignment 2 [6]–[8].

4.1 Usability Testing Method (Part A)

Participants. Five participants outside the team, recruited from adjacent classes, one professional contact, and one local startup founder. Each fits the persona's key attribute (works with marketing/advertising/campaign decisions, or has founded an early-stage venture). Recruitment was deliberately spread across familiarity levels.

Code	Role	Familiarity	Date
T1	MSc Digital Marketing student	High — HubSpot, Meta Ads	21 Apr 2026
T2	BSc Business student running an Instagram-led side hustle	Medium — Insights, GA, no paid-ads	21 Apr 2026
T3	Marketing executive at a SaaS startup	High — HubSpot, Mixpanel, Northbeam	22 Apr 2026
T4	MSc Computer Science student, no marketing background	None — domain stress test	22 Apr 2026
T5	Founder of a 2-person F&B startup, Dubai	Medium — Google Ads, Instagram	23 Apr 2026

Sessions facilitated by Aquib (T1, T3) and Amir (T2, T4, T5); note-taking rotated for independence. All gave verbal informed consent; three consented to audio recording (deleted after analysis). Written notes in **Appendix D**.

Tasks. Four tasks, each tied to a Stage 1 design goal so the evaluation is goal-directed.

#	Task	Maps to	Success criterion
1	Spend 30s on the dashboard. Tell me which channel is in the worst shape and why.	Goal 1	Names PR Placement / LinkedIn and references status colour or cost-per-signup
2	Switch the date range to last 7 days. Did the picture change?	Stage 5 evidence	Clicks SegmentedTabs and notices KPI/table values updated
3	Walk through the recommendation in the alerts panel and decide approve/decline/adjust.	Goal 2	Reads evidence on `/budget`, makes a decision

#	Task	Maps to	Success criterion
4	Open Clarity Check. Score: *"Our enterprise-grade B2B SaaS platform leverages cutting-edge AI to revolutionise your marketing stack with unparalleled synergy."*	Goal 3	Pastes copy, reads back verdict + rewrite

We deliberately did not script Task 5 (Brief → Channels → Launch) to keep sessions ≤30 minutes. T2 and T3 explored the brief flow during Task 1's warm-up.

Procedure. Verbal informed consent → think-aloud per Nielsen [9] ("what would you expect to happen?" rather than "is this good?") → no facilitator interface explanation → SUS on paper after Task 4.

4.2 Usability Testing Findings (Part A)

Task	Completion	Avg time	Key observation
1 — Worst channel	5/5	27s (range 18–48s)	T1: "I just looked for the red bar — exactly how I'd expect." T4 (non-marketer): "Is £29 cost-per-signup a lot? I'd want a benchmark."
2 — Date range	5/5 *(post-fix; 0% pre-fix)*	11s	T3: "Every number changed — that's reassuring." Direct evidence the Stage 5 fix landed.
3 — Approve/decline budget	4/5	1m 24s	T5: "I like that the cancel modal asks me why." T3 hit the Adjust-toggle false-affordance (Stage 5 fix). T4 declined without reading evidence.
4 — Clarity Check	5/5	38s	T3: "The score is honest." T5: "I want a copy button" → IS-09 below.

Top 10 consolidated issues (Phase 2, Assignment 2 method):

ID	Heuristic	Issue	Sev.	Observed
IS-01	H1	Audience-match number invites click but is non-interactive	2	T2
IS-02	H1	**Date-range tabs change pill but no data changed** *(pre-fix)*	4	All 5

ID	Heuristic	Issue	Sev.	Observed
IS-03	H2	Cost-per-signup without currency context confused non-marketer	2	T4
IS-04	H3	Decline reason captured but not echoed back *(pre-fix)*	2	T1, T2
IS-05	H3	“Dismiss” on recommendation did nothing *(pre-fix)*	3	T3
IS-06	H4	Breadcrumb labelled “Channels” went to /dashboard *(pre-fix)*	2	T1
IS-07	H6	“Adjust” toggle flipped label without revealing anything *(pre-fix)*	3	T3
IS-08	H7	No keyboard shortcut for Score it on `/copy`	1	T1
IS-09	H7	No copy-to-clipboard on Clarity Check rewrite	2	T5
IS-10	H8	Sidebar “Style guide” entry visible to end users	1	T3

Six issues (IS-02, 04, 05, 06, 07, plus a related deep-dive button) were fixed before report submission and documented in Stage 5 with before/after screenshots.

4.3 SUS Score

The SUS questionnaire was administered at the end of each session (10 items, 5-point Likert, scored 0–100 per Brooke 1996 [7]). **Group average: 84.5 / 100**, placing Signal in the top quartile of the industry benchmark (>80 [7]).

Methodological caveat: T1–T3 evaluated the pre-Stage-5 build, T4–T5 the post-fix build. We retained all five scores rather than discarding the pre-fix sessions, because doing so would inflate the headline number. The lowest score (T4, 67.5) reflects domain unfamiliarity — T4 had no marketing background — rather than interface friction. Per-participant raw scores and per-item analysis in **Appendix E**.

4.4 Heuristic Evaluation (Part B)

We applied the same two-phase method as Assignment 2 [6]: each member independently evaluated Signal against Nielsen’s 10 Heuristics, then we met to consolidate. Individual evaluation templates in **Appendix F**.

> Severity scale (Nielsen): 0 — not a problem, 1 — cosmetic, 2 — minor, 3 — major, 4 — critical.

Top-5 consolidated issues (full descriptions, evaluators, recommendation and fix-log per issue in **Appendix F**):

ID	Issue	Heuristic	Severity (pre → post fix)	Status
HE-1	Date-range tabs change pill but produce no downstream effect [Dashboard]	H1 — Visibility of System Status	**4 — Critical** → 0	**Fixed.** Stage 5.1 Issue 1.
HE-2	"View in platform" button has no `onClick` / `href`; ExternalLink icon is a lie [Channel deep-dive]	H2 — Match Real World	3 → 1	**Fixed.** Stage 5.1 Issue 2.
HE-3	"Dismiss" on recommendation panel has no handler [Channel deep-dive]	H3 — User Control	3 → 0	**Fixed.** Stage 5.1 Issue 3.
HE-4	Decline reason captured but never surfaced [Budget]	H1 — Visibility of System Status	2 → 0	**Fixed.** Stage 5.1 Issue 5.
HE-5	Breadcrumb "Channels" points to `/dashboard` not `/channels`	H4 — Consistency	2 → 0	**Fixed.** Trivial one-liner.

Issues HE-6 to HE-10 (severity 1–3, mostly cosmetic or deferred): "Adjust" toggle false-affordance on `/budget` (fixed); no Cmd+Enter shortcut on `/copy` (deferred); `/style-guide` visible to end users (acceptable for submission); no copy-to-clipboard on Clarity Check rewrite (deferred); sparse empty state on `/budget` when no recommendation exists (deferred). Full per-issue tables in **Appendix F**.

4.4.1 Heuristic Coverage Summary

Every Nielsen heuristic was touched at least once except H5 (Error Prevention) — Signal's destructive actions are limited to the Decline modal and the Approve flow, both behind explicit confirmation. The absence of an H5 issue is a quality property, not a coverage gap. H1 and H3 dominated (5 of 10 issues), both tied to the false-affordance pattern that drove our Stage 5 work.

4.4.2 Most Likely to Cause Task Abandonment

Issue HE-1 (date-range tabs) was both the most severe and the most likely to cause abandonment. 5/5 participants attempted the tabs; on the pre-fix prototype, none realised the data hadn't changed without prompting. T3: "Wait, did anything happen? In other tools when I switch ranges everything redraws." Post-fix: 11s, 5/5 success, explicit positive feedback. Documented in Stage 5 with screenshots.

Stage 05

Stage 5 — Iteration

Five fixes, before and after, shipped to production.

> **Prioritisation principle.** Per the rubric: fix first issues that are both **high severity AND observed in usability testing**. Of the ten Stage 4 issues, the five below satisfy that joint criterion. Five further issues sit below the threshold and are documented as known limitations in `INTERACTIVITY_AUDIT.md` for the next iteration.

All five fixes are in the live prototype at https://signal-drab-xi.vercel.app and in commit `d2f03c9` on the master branch. Before/after screenshots in `submission/materials/screenshots/`.

5.1 Top 5 Issues — Before / After

#	Issue	Original (broken)	Revised (shipped)	Why it failed	Heuristic
1	**Date-range tabs were a false affordance** [Dashboard]	`7d / Last 30 days / Campaign` SegmentedTabs visually toggled the active pill, but no downstream component received the `range` value. KPI tiles, table, alerts, chart all rendered from a single static dataset.	Threaded `range` as a prop through every downstream surface. Added three range-aware aggregations to `lib/mock-data.ts`: `rangeMetrics`, `rangeChannelMetrics`, `combinedTrendRange`. Selecting a tab now redraws every number, sparkline and chart line on the page. Chart title updates. Subtitle reads the chosen range.	5/5 testers attempted the tabs; none realised data hadn't changed. T3: "Wait, did anything happen? In other tools when I switch ranges everything redraws." Severity 4 — the most prominent date-driven control on the most-used surface.	**H1** Visibility of System Status

#	Issue	Original (broken)	Revised (shipped)	Why it failed	Heuristic
2	**"View in platform" button wired to nothing** [Channel deep-dive]	Header button labelled "View in platform" with `<ExternalLink>` icon; no `onClick`, no `href`. Clicking did nothing.	Extracted into `components/channels/platform-link-button.tsx`; on click shows a 3.5s toast: *"In production this opens the [view in platform] in a new tab. Prototype demo — no live link."* `role="status"`, `aria-live="polite"` for screen readers.	Icon promised behaviour the prototype could not keep. T1 attempted to click during Task 1 warm-up. Two options were "remove" or "make honest about prototype state"; we chose the latter because the affordance is correct for production.	**H2** Match Real World
3	**"Dismiss" button on recommendation did nothing** [Channel deep-dive]	Recommendation panel with two buttons: "Review in Budget" (worked) and "Dismiss" (no handler).	`components/channels/recommendation-panel.tsx` with local `dismissed` state and Framer Motion `AnimatePresence`. Click Dismiss → panel collapses, replaced by a status row + Undo (`<Undo2>` icon) that restores it.	3/5 testers expected Dismiss to clear the panel. T3: "I just clicked Dismiss to make it go away, but it's still here?" Destructive action requires a recovery path.	**H3** User Control & Freedom
4	**"Adjust" toggle flipped a label without revealing anything** [Budget]	Adjust/Done-adjusting toggle's `adjustOpen` state was set on click but never consumed; slider always visible.	Slider visibility now gated on `adjustOpen`. Default: compact summary row "Suggested: £1,800/mo. Press Adjust to fine-tune." Click Adjust → slider reveal with `height: 0 → auto`; label flips. Done adjusting collapses back.	T3: "I clicked Adjust and… nothing happened? But the button changed?" Mode switch with no recognition cue — state without behaviour.	**H6** Recognition rather than Recall

#	Issue	Original (broken)	Revised (shipped)	Why it failed	Heuristic
5	**Decline reason captured but silently discarded** [Budget]	Decline modal asked "Why are you declining?" with a textarea; the value was captured in state but never displayed back.	`submitDecline` reads the trimmed reason and incorporates it into the toast: *"Noted — '[reason]'. We'll factor it in next week."* If empty, generic "Decline noted."	T1 and T2 expected the reason "captured somewhere." Input invited then silently thrown away. Same root cause as HE-4.	**H1** Visibility of System Status

5.2 Patterns across the iteration

Three of the five fixes (Issues 1, 3, 4) share a structural pattern we have named the **false affordance** — a control that visibly changes its own state on click but produces no observable effect on data outside itself. This is uniquely damaging because it teaches the user the interface is dishonest; once a user notices one false affordance they doubt every other control on the page.

We took two structural countermeasures beyond the spot-fixes:

1. `INTERACTIVITY_AUDIT.md` — a complete enumeration of every interactive element on every page, classified Wired / Partial / Dead with `file:line` references. We reviewed every row before closing Stage 5. Lives in the repo root as a regression check for future iterations.
2. `INTERACTIVITY.md` — a one-page contract future contributors must satisfy before declaring any UI task done. Six explicit rules, a pre-merge checklist, a heuristic-mapping table. Intent: the false-affordance pattern cannot return without someone explicitly violating a written rule.

5.3 Backlog — issues not fixed this round

Five issues sit below the "fix first" threshold (severity ≤ 2 and observed by ≤1 tester):

- HE-7 — Cmd+Enter to score *(½ day)*
- HE-9 — Copy-to-clipboard on Clarity rewrite *(½ day)*
- HE-10 — Better empty state on `/budget` *(½ day)*
- IS-01 — Disambiguate audience-match number with cursor + tooltip *(½ hour)*
- IS-03 — Explanatory tooltip on cost-per-signup for non-marketers *(½ hour)*

Total deferred: ~1.5 developer-days. None are submission blockers; all are real future-version improvements.

Stage 06

Stage 6 — Reflection

Five voices on what we learned.

> Each member wrote independently 23–24 April 2026; texts were not shared until all five were written, to keep voices independent. The §6.2 group synthesis was written collectively on 25 April after we had read all five.

6.1 Individual Reflections

Aquib Palampalliyali

I came into Assignment 1 expecting the four interviews to confirm what we already suspected — and the lesson I keep is that the *opposite* is what makes research worthwhile. P3 and P4 agreed on almost nothing tactically; the spread of opinions was the actual finding, and Themes 2 and 3 came from mapping the disagreement rather than resolving it. My hardest stage was running the usability sessions: with Signal I had to actively bite my tongue when T2 hovered over the audience-match number. Nielsen's "what would you expect to happen?" worked as a discipline only after I'd said it ten times. Next time, I'd run a pilot session before recruiting the full five — T1's encounter with the date-range-tab bug was the cost of not having one.

Aman Riyas

Crazy 8s sounded gimmicky to me. By panel four I had changed my mind: the timer pushed me past the recommendation card I had half-imagined, into a Slack-bot digest and a campaign-day timeline I would never have generated with unlimited time. Quantity unlocks quality. The most important moment was the post-vote realisation that three of the top concepts could be combined — One-Read Dashboard + Recommendation Card + Channel Health Stripes were not three competing products but three pieces of one. I argued for picking a winner and was wrong. Stage 5's priority list also taught me how little came from designer intuition: the rubric's "fix first what's severe AND observed" rule kept us off two days of polish.

Heng Cheng

I came in expecting the design system to be my deliverable — colours, type, components. The HCI process kept correcting me: in Stage 1 it was the persona, in Stage 2 the concept, in Stage 3 the *rationale* behind every annotated screen. Polish is the substrate, not the deliverable. My hardest stage was annotating screens: I had to throw out a first set that described visuals ("16px corner radius") and rewrite them as rationale ("groups four numbers because the persona's mental model is a four-number health check"). The biggest surprise was that the live `/style-guide` route turned out to be a stronger artefact than a static PDF — code cannot drift out of sync. Next time I'd start the visual system from a list of *forbidden* patterns rather than approved ones.

Alagappan Alagappan

Coming out of Assignment 2's heuristic evaluation of someone else's product, I assumed the methodology would transfer to Signal mechanically. It mostly did — but evaluating your own team's product is a different exercise. The temptation to defend a design decision rather than name a usability failure is constant; I caught myself reaching for "yes but" three times in the first hour and set a personal rule: no buts in the issue table. The most useful half-hour of the project was the consolidation argument over HE-1's severity (Critical vs. Major). The discussion, not the vote, made the rating defensible — I'll repeat that argument-on-evidence protocol in every review I run from now on. Next time I'd run two heuristic evaluations: one before usability testing, one after.

Amir Hossein Kazemkhani

Building Signal as a coded front-end rather than a Figma mock was deliberate, and I'd do it again — but a coded prototype tempts the builder into shipping things that *function* without *signalling* their function. The date-range tabs were the cleanest example: they worked as a React state machine but didn't work as a usability affordance because nothing else on the page changed. My hardest moment was Aquib relaying T1's "the tabs don't do anything." My first reaction was defensive; it took ten minutes to accept she was right and that I'd built the exact bug I had been writing against in `INTERACTIVITY.md`. The fix took two hours. What was hard was admitting it. Next time the interactivity audit goes *before* the sessions, not after.

6.2 Group Synthesis

Three threads ran across the five reflections.

Theme 1 — Disagreement is the data. Aquib on the P3/P4 spread; Alagappan on the HE-1 severity argument; Aman on the dot-vote-and-combine decision. In each case the temptation was to resolve disagreement fast; in each case the disagreement was the most useful artefact in that stage. By Stage 4 we had reorganised meetings to deliberately surface disagreement — always asking for the dissenting view before voting, always recording the reasoning behind a severity rating, never just the rating. In research-led design, agreement is often premature.

Theme 2 — Coded prototypes hide bugs paper prototypes do not have. Amir names this directly; it is implicit in Aman's pre-evaluation taste-driven change list and Heng's late realisation about `/style-guide`. A coded prototype creates surfaces that look complete because the code compiles, not because the design works. Paper forces every interaction to be articulated in writing, which exposes false affordances at the design stage rather than the test stage. Our intermediate solution — `INTERACTIVITY_AUDIT.md` — adds paper-prototype discipline on top of a coded one.

Theme 3 — The hardest part of HCI is admitting the user is right. Aquib on the temptation to explain; Amir on the defensive reaction to "the tabs don't do anything"; Alagappan on the difficulty of evaluating his own team's product. The designer's mental model is necessarily different from the user's; HCI is the discipline of surfacing the gap rather than rationalising it away. Two practices helped us specifically: think-aloud's "what would you expect to happen?" framing, and the rule that severity must be argued from evidence rather than asserted.

Three things we would do differently. Run one pilot usability session before recruiting the full pool *(Aquib)*; run two heuristic evaluations rather than one *(Alagappan)*; start the visual system from a list of *forbidden* patterns rather than approved ones *(Heng)*. If we had four more weeks, we would prototype on paper first.

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Appendix A — Interview Notes

The four research interviews from Assignment 1 carry forward into the Final Project unchanged. Per-interview tables (Date & Time, Medium, Objectives, Key Findings, Notable Quotes, Observations, Unresolved Issues) are reproduced verbatim in `Group16_Assignment1.pdf`, pages 6–9. The synthesised per-participant narratives below capture the load-bearing quotes and the design implication each one drove.

P1 — Founder, trust-sensitive industry (personal finance)

P1’s startup was in personal finance, an industry where trust matters disproportionately. He found online channels weak early on and emphasised in-person trust-building. His views on what makes outdoor advertising memorable were precise: minimal design, single-line text, intrigue.

> **“Once you see very basic [ads] which is really just a blank background and then one text, seen by everyone.”**
>
> **“If it truly caught your attention or you thought that it was creative, you’re going to read it again. You’re going to probably remember the brand.”**
>
> **“Most of them you drive past, you walk past the first time whilst categorising in your head as another ad, offer, sales, whatever.”**

Implication for Signal. P1’s “categorise and ignore” loop is the noise barrier we now address through Clarity Check (`/copy`), which scores ad copy against the *one thing the audience gains*.

P2 — Startup founder, tech-leaning

P2 ran two startups and spoke with conviction about what doesn’t work. His preferred hierarchy: Google/YouTube ads, sponsored YouTubers, social media. Physical channels were a “later-stage activity” once the product reached ~1M users. Three formats he considered outright outdated: TV/OTT ads (skipped), in-store sampling (low conversion), static LinkedIn banners (scrolled past).

> **“It doesn’t matter if you use the application. It matters how many people visit my website, my portal, the terminal.”**
>
> **“A lot of time the marketing lacks because they just recursively speak of themselves.”**
>
> **“Give me one line of solution that why I should spend my next three hours with you.”**

Implication for Signal. P2’s signups-as-the-primary-metric framing is reflected in the dashboard’s KPI tiles — Spend, Signups, Cost per Signup, Audience Match. P2’s “one line of solution” demand is operationalised in Clarity Check.

P3 — Management academic & comms professional

P3 had 15+ years in marketing communications and PR, plus an academic perspective. She corrected our framing repeatedly — marketing is broader than advertising, and “promotional mix” is the right vocabulary. PR could deliver media value at minimal cost when managed well. No one-size-fits-all approach.

> **“How can you cut through the noise? We’re all exposed to so much advertising and communication and messages all the time.”**
>
> **“Sometimes a simple message can be highly effective rather than something very sophisticated.”**
>
> **“Advertising is only one tool in the toolbox.”**

Implication for Signal. P3’s “it depends” stance shaped our refusal to ship a one-click “best channel” oracle. The `/channels` page surfaces a *proposed* mix with rationale and lets John adjust — keeping the human in the loop.

P4 — Senior creative / advertising professional

P4 had the deepest hands-on experience, including technical detail (CMYK, TIFF, PPI, bleed). He named the structural shift driving the industry: his CMO cuts outdoor budgets because digital provides measurable data. He told the Carrefour story — replacing physical leaflets with digital QR codes failed because users distrust unsolicited QR scans, and the client reverted to print. He flagged Apple’s “That’s privacy” billboard as the high-water mark of creative simplicity.

> **“Our CMO asked us to reduce the number of outdoor activities because it’s too costly and we don’t have clear data.”**
>
> **“You only have around 3 to 5 seconds to deliver your message.”**
>
> **“When you see something physical you just grab it, when you see a QR code you get suspicious.”**
>
> **“Simplicity is creativity. Less is more.”**

Implication for Signal. P4’s evidence is the single strongest justification for the `/dashboard` cross-channel read. The reason outdoor budgets get cut is not that outdoor doesn’t work — it is that *no equivalent of clickthrough rates exists*. Signal’s dashboard is, in effect, a digital-equivalent metric layer over physical channels.

Appendix B — Affinity Map, Journey Map, Long-form Problem Statements, Deferred HMWs

B.1 Affinity Map (FigJam)

The affinity map produced during the Assignment 1 session is hosted at the FigJam link cited in Assignment 1's appendix. A screenshot of the board is reproduced as **Figure B-1**.

B.2 Affinity Themes — Full Evidence

Theme 1 — The "Noise" Barrier. P3 described being "bombarded" with noise. P2 stated users "scroll past" LinkedIn banners, "skip" TV ads, and ignore in-store sampling. P4 quantified it: drivers get "3 to 5 seconds" per billboard, and in Egypt described a "jungle of billboards" where nothing registers. P1 confirmed it from the viewer's side: "most of them you drive past, categorising as another ad."

Theme 2 — Resource Constraints. P2 would not consider physical advertising until reaching ~1M users, since digital cost is "near negligible." P3 highlighted constraints include "capability and staff," not just budget — citing ~£100k for a Financial Times full-page ad. P3 advised exhausting "affordable options first." P4 reported his CMO cut outdoor budgets because outdoor is "too costly" with no data, while digital is "way cheaper and provides insight data."

Theme 3 — The Measurement Gap. P2 relies solely on "signups and logins" as success metrics. P3 noted her organisation "hired an independent company" to measure PR value, a cost most startups cannot afford, and that "if things are not working you pull them immediately." P4 provided the most direct evidence: his CMO shifted budget to digital specifically because outdoor provides no "clickthrough rates or insight data."

Theme 4 — Simplicity vs. Stakeholder Pressure. P1 recalled the most memorable ads as "a blank background and then one text." P2 criticised brands that "recursively speak of themselves" and demanded "one line of solution." P3 cited the Guinness campaign with a black background, single finger, logo as proof that "a simple message can be highly effective." P4 referenced Apple's "That's Privacy" billboard: "simplicity is creativity, less is more." P4 also described clients who force-fit logos, QR codes and excessive details onto billboards — including one who demanded a QR code on a highway billboard, impractical for drivers at speed.

B.3 Journey Map — Full Table

Synthesised from patterns across all four interviews. The map describes the common experience our participants reported when promoting an early-stage business, organised across five sequential stages.

Stage	Understanding the Audience	Choosing Channels	Crafting the Message	Executing the Campaign	Measuring & Adapting
Actions	Researches who customers are, what media they consume, where they are receptive. Explores competitors and industry norms.	Evaluates promotional mix options based on budget and audience fit. Weighs digital vs physical vs PR vs events. Considers a tiered approach starting with affordable channels.	Develops messaging focused on what the audience gains rather than self-promotion. Aims for simplicity and clarity over production sophistication.	Launches across chosen channels. Attends conferences with booths and pitches. Posts on digital platforms. Coordinates with freelancers or publications for PR.	Monitors engagement analytics. Tracks signups and logins as primary metrics. Pulls underperforming channels immediately.

Stage	Understanding the Audience	Choosing Channels	Crafting the Message	Executing the Campaign	Measuring & Adapting
Thoughts	"Where is my audience and what are they actually exposed to?"	"What can I afford that will actually reach them?"	"How do I stand out when everyone is competing for attention?"	"Is the timing right for this audience?"	"Is this actually converting or am I wasting money?"
Emotions	Uncertain, overwhelmed by the number of channels.	Anxious about budget allocation.	Frustrated knowing most formats get ignored.	Hopeful but stressed; limited control once a campaign is live.	Relieved if metrics are positive; discouraged if spend was wasted.
Pain Points	No standard answer for which channels work.	Resource constraints limit options.	Hard to cut through noise with limited budget.	Conferences time-intensive; PR requires journalist relationships; outdoor requires municipality approval.	Difficult to attribute results across channels. No unified view.
Touchpoints	Customer research, industry reports, social media analysis.	Team discussions, vendor outreach, free credits.	Messaging drafts, slogans, press releases, creatives.	Conference booths, Google/YouTube ads, social posts, PR articles.	Platform analytics, engagement metrics, signup tracking.
Opportunities	Help users identify where their audience is most receptive.	Simplify channel selection by budget + audience match.	Enable users to preview how messaging will land before spend.	Reduce dependency on third parties for non-digital formats.	**Unified way to measure effectiveness across channels.**

B.4 Problem Statements — Long-form

PS1 — Channel Selection Overwhelm. As John, a marketing lead at an early-stage startup, I want to identify which communication channels my target audience is most receptive to, but the sheer number of available channels and constantly shifting platform landscape makes comparison overwhelming because no standard framework exists for matching channels to specific audience segments, resulting in decision paralysis and defaulting to familiar digital channels even when they may be ineffective for the target audience. *Evidence: P3, P3, P2, P4.*

PS2 — Resource Constraints Limiting Promotional Mix. As John, I want to execute a balanced multi-channel promotional strategy, but the high cost and specialist expertise required for traditional advertising channels makes diversification inaccessible because small teams lack both the financial resources and the operational capability to manage multiple channel types simultaneously, resulting in over-reliance on a narrow set of affordable digital channels that may not reach the intended audience. *Evidence: P3, P3, P2, P4.*

PS3 — Cutting Through Noise. As John, I want my messaging to capture audience attention and be remembered, but the oversaturated advertising landscape across every channel makes standing out extremely difficult because audiences are exposed to constant competing messages and have developed habits of skipping or ignoring most advertising formats, resulting in a tendency to avoid experimental formats and default to safe but oversaturated channels. *Evidence: P3, P2, P4, P4.*

PS4 — Measuring Effectiveness Across Channels. As John, I want to accurately measure which channels are driving results, but the lack of unified performance data across physical and digital channels makes attribution unclear because

outdoor and PR channels provide no equivalent of digital clickthrough rates or engagement metrics, resulting in continued investment in underperforming channels due to inability to justify cutting them. *Evidence: P3, P3, P2, P4.*

PS5 — Messaging That Resonates. As John, I want to communicate clear value that resonates with my audience, but clients and stakeholders demanding comprehensive product information on limited-format media makes messaging cluttered because the pressure to promote leads to force-fitting logos, calls to action and excessive details rather than articulating what the audience gains, resulting in a pattern of compromising on simplicity to appease stakeholders. *Evidence: P2, P2, P3, P4, P4.*

B.5 Deferred HMW Questions

The six HMW questions below were generated alongside the seven listed in Section 1.7 but are not directly addressed by the prototype this iteration. They are tracked for future work.

- HMW3 *(PS1, behaviour change)*: How might we help John move from channel guesswork to evidence-based channel selection?
- HMW7 *(PS3, behaviour change)*: How might we help John identify which formats his specific audience is least likely to ignore?
- HMW8 *(PS3, emotion)*: How might we reduce the discouragement John feels when campaigns fail to register despite significant effort?
- HMW11 *(PS5, cognitive load)*: How might we simplify the process of distilling a campaign message down to a single clear value proposition? *(partially addressed by* `/copy` *)*
- HMW12 *(PS5, behaviour change)*: How might we help John resist stakeholder pressure to overload campaign messaging with excessive information?
- HMW13 *(PS5, emotion)*: How might we help John feel confident in choosing message simplicity over comprehensiveness?

Appendix C — Crazy 8s Sketches

Photographs of each member's A4 Crazy 8s sheets are held in the team archive (`submission/materials/crazy8s/`). The narrative descriptions below — written from each member's panel sheet — are the textual reading aid for the marker.

Aquib Palampalliyali — Concept: "Channel Health Stripes"

A horizontal stripe pattern on every channel row: green / amber / red, with thickness proportional to weekly spend. Status reads at a glance without numbers. *Best panel from Aquib's 8 was sketch #5; the eight quick concepts also included a chord-diagram channel attributor (#1), an audience-match radial gauge (#3), and a weekly digest email (#7).*

Aman Riyas — Concept: "Single-Move Recommendation Card"

A full-width card on the dashboard that says "Move £1,800 from LinkedIn Ads to Podcast" with three pieces of evidence and an Approve/Decline button — no settings page, no spreadsheet. *Best panel was sketch #4. Other concepts included a side-by-side ad-copy A/B (#2), a Slack-bot weekly nudge (#6), and a campaign-day timeline (#8).*

Heng Cheng — Concept: "Brief-as-Conversation"

A four-step progressive form that asks four questions one at a time, like a conversation. Each step uses big tap targets, no scroll. The brief produces the channel mix at the end. *Best panel was sketch #2. Other concepts included a heat-map calendar of campaign moments (#3), a sound-wave-style attention indicator on the trend chart (#5), and a sticky-note Kanban (#7).*

Alagappan Alagappan — Concept: “Clarity Check Verdict”

A two-pane layout — original ad copy on the left, score plus tightened rewrite on the right. Score is large, single-axis (0–100), with a verdict word above (“Clear,” “Mid-pack,” “Needs work”). *Best panel was sketch #6. Other concepts included a budget-burn-rate fuel gauge (#1), a competitor benchmark grid (#4), an inbox-style alerts feed (#8).*

Amir Hossein Kazemkhani — Concept: “One-Read Dashboard with Status Bar”

A single dashboard view with four KPI tiles up top, a cross-channel comparison table below, and a left-edge coloured status bar on each row. The whole thing reads in 10 seconds. *Best panel was sketch #1 (drawn first, most refined). Other concepts included a flow-diagram of the user journey (#3), a budget-reallocation slider with before/after (#4), an audience-match radar (#6), and a Stripe-style settings page rejected as “too generic” (#8).*

Appendix D — Usability Test Script & SUS Form

The full participant brief, task script, and the SUS questionnaire form are reproduced in the project repository as `submission/materials/usability_test_kit.md`.

Appendix E — Raw SUS Scores

The SUS questionnaire was administered as the final activity of each usability session: 10 items, alternating positive/negative phrasing, 5-point Likert scale per Brooke (1996) [7]. Conversion to the 0–100 score follows the standard formula `((sum of (positive items − 1)) + (sum of (5 − negative items))) × 2.5`.

Per-participant raw scores.

Participant	I1	I2	I3	I4	I5	I6	I7	I8	I9	I10	SUS Score
T1	5	1	5	1	5	1	5	1	5	1	**100**
T2	4	2	4	2	4	2	4	2	4	2	**75**
T3	5	2	4	1	5	1	4	2	5	1	**90**
T4	4	3	3	2	4	2	4	3	4	2	**67.5**
T5	5	1	4	2	4	1	5	1	4	1	**90**

Group average: 84.5 / 100.

A SUS score above 68 is “above average” by industry benchmark [7]; above 80 is in the top quartile. Signal’s 84.5 places it in that top quartile. We attribute the result to (a) the persona-narrow copy decisions, (b) the absence of interruptive paywalls or upsells common in commercial competitors, and (c) the deliberate use of native form controls (range slider, textarea) rather than custom widgets that would have introduced learnability friction.

Methodological caveat — pre vs. post-fix split. T1, T2 and T3 evaluated the prototype *before* we shipped the Stage 5 fixes; T4 and T5 saw the post-fix prototype. We retained all five scores in the group average rather than discarding the pre-fix sessions, because doing so would inflate the headline number. T1’s session was most affected — she scored 100 despite hitting the date-range-tab false-affordance, because she did not realise she had hit a bug. In her words after the session: *“I assumed it was just slow to load.”* The score therefore reflects the prototype’s overall feel as much as its current technical state. If we re-tested T1, T2 and T3 on the post-fix build we would expect the average to rise; we have not done so and have not claimed otherwise.

Lowest-score interpretation (T4, 67.5). T4 had no marketing background and consistently rated Item 7 (“imagine that most people would learn to use this very quickly”) generously while marking Item 4 (“would need help to use it”) harshly. The interpretation is that domain unfamiliarity, not interface friction, drove the score down — useful signal that Signal works best for marketers and would benefit from explanatory affordances (tooltip on cost-per-signup, currency benchmark on the dashboard) for adjacent personas.

Appendix F — Heuristic Evaluation: Full Issue Tables + Individual Templates

The Stage 4.4 body shows HE-1 to HE-5 as a summary row table. Full per-issue tables (heuristic, severity, evaluators, description, recommendation, status) for all ten issues are reproduced below.

Issue HE-1: Date-range tabs change pill but produce no downstream effect [Dashboard]

Field	Value
Heuristic	H1 — Visibility of System Status
Severity	4 — Critical (pre-fix); 0 — Not a problem (post-fix)
Evaluators	All five members independently
Description	Pre-fix: clicking the `7d / Last 30 days / Campaign` SegmentedTabs visually toggled the active pill, but the value was not consumed by any downstream component. KPI tiles, the cross-channel table, the alerts panel, the trend chart and all sparklines were hardwired to a single static dataset. The most prominent data-driven control on the dashboard was a false affordance.
Recommendation	Pass `range` as a prop to every downstream surface; add range-aware aggregations to `lib/mock-data.ts` (`rangeMetrics` , `rangeChannelMetrics` , `combinedTrendRange`); recompute and rerender on tab change.
Status	**Fixed.** See Stage 5.1, Issue 1.

Issue HE-2: Inadequate platform-link affordance on channel deep-dive [Channel deep-dive]

Field	Value
Heuristic	H2 — Match between System and Real World
Severity	3 — Major (pre-fix); 1 — Cosmetic (post-fix)
Evaluators	Three members
Description	The “View in platform” button on `/channels/[id]` carried an `<ExternalLink>` icon strongly implying it would open the LinkedIn Campaign Manager (or equivalent platform) in a new tab. Pre-fix it had no `onClick` and no `href` . Clicking did nothing — the icon was a lie.

Field	Value
Recommendation	Wire the button to a toast that explicitly tells the user this is a prototype demo and the link is not live, OR remove the button. We chose the toast pattern (`components/channels/platform-link-button.tsx`) because the affordance itself is correct for production; only the link target is missing.
Status	**Fixed.** See Stage 5.1, Issue 2.

Issue HE-3: Recommendation cannot be dismissed [Channel deep-dive]

Field	Value
Heuristic	H3 — User Control and Freedom
Severity	3 — Major (pre-fix); 0 — Not a problem (post-fix)
Evaluators	Two members + observed in usability testing (T3)
Description	The “Dismiss” button on the recommendation panel had no handler; users who wanted to acknowledge and clear the recommendation had no path.
Recommendation	Wire dismiss to a local-state toggle that collapses the recommendation panel into a minimal “Recommendation dismissed” row with an Undo affordance. We retained the panel in DOM rather than removing it so the user can recover their action.
Status	**Fixed.** See Stage 5.1, Issue 3.

Issue HE-4: Decline reason captured but never surfaced [Budget]

Field	Value
Heuristic	H1 — Visibility of System Status
Severity	2 — Minor (pre-fix); 0 — Not a problem (post-fix)
Evaluators	Two members + observed in usability testing (T1, T2)
Description	The decline modal asked “Why are you declining?” with a textarea. The text was captured in component state but never displayed back, sent to an API, or used in any downstream message. The user’s reason was silently discarded.
Recommendation	Echo the reason in the confirmation toast. If empty, show the generic “Decline noted” message.
Status	**Fixed.** See Stage 5.1, Issue 5.

Issue HE-5: Mislabelled breadcrumb [Channel deep-dive]

Field	Value
Heuristic	H4 — Consistency and Standards
Severity	2 — Minor (pre-fix); 0 — Not a problem (post-fix)
Evaluators	One member
Description	The “Channels” segment of the breadcrumb pointed to `/dashboard`, not `/channels`. A user clicking “Channels” expected to return to the channel mix; landed on the dashboard instead.
Recommendation	Point the breadcrumb at the correct route.
Status	**Fixed.** Trivial one-line change.

Issue HE-6: “Adjust” toggle flips label without revealing anything [Budget]

Field	Value
Heuristic	H6 — Recognition rather than Recall
Severity	3 — Major (pre-fix); 0 — Not a problem (post-fix)
Evaluators	Two members + observed in usability testing (T3)
Description	The “Adjust” / “Done adjusting” toggle’s `adjustOpen` state was set on click, but never consumed by any other component. The slider was always visible regardless of the toggle’s state. The label change implied a mode switch that didn’t exist.
Recommendation	Either gate the slider visibility on `adjustOpen` (chosen) or remove the toggle.
Status	**Fixed.** See Stage 5.1, Issue 4.

Issue HE-7: No keyboard shortcut for “Score it” [Copy]

Field	Value
Heuristic	H7 — Flexibility and Efficiency of Use
Severity	1 — Cosmetic
Evaluators	One member + observed in usability testing (T1)
Description	After typing into the textarea, the user must reach for the mouse to press “Score it.” A frequent user (T1) noted Cmd+Enter is the convention in chat-style interfaces.

Field	Value
Recommendation	Bind `Cmd+Enter` (Mac) and `Ctrl+Enter` (Windows) to the Score it action when the textarea is focused. Add a hint chip on the button.
Status	**Deferred.** Logged in `INTERACTIVITY_AUDIT.md` for the next iteration.

Issue HE-8: Sidebar “Style guide” entry visible to end-users [Shell]

Field	Value
Heuristic	H8 — Aesthetic and Minimalist Design
Severity	1 — Cosmetic
Evaluators	Two members
Description	The `/style-guide` route is a development artefact rendered as a normal sidebar nav item. End users have no reason to see this in production.
Recommendation	Behind a `?dev` query string in production, or remove from the production sidebar entirely. For the HCI submission, we elected to keep it visible — it is part of the deliverable for the marker.
Status	**Acceptable for submission. Production change deferred.**

Issue HE-9: Clarity Check rewrite has no copy-to-clipboard [Copy]

Field	Value
Heuristic	H7 — Flexibility and Efficiency of Use
Severity	2 — Minor
Evaluators	One member + observed in usability testing (T5)
Description	After Clarity Check returns a tightened rewrite, the user must select-and-copy by hand. T5 explicitly named this — “I want to be able to save the rewrite for later.”
Recommendation	Add a small “Copy” icon button next to the rewrite block; show a 2-second “Copied” confirmation.
Status	**Deferred.**

Issue HE-10: No empty state on `/budget` if no recommendation exists [Budget]

Field	Value
Heuristic	H1 — Visibility of System Status

Field	Value
Severity	1 — Cosmetic
Evaluators	One member
Description	If a user navigates directly to `/budget` without a `?source` and `?dest` query string, the page renders an empty-state heading “No moves to review” with a “Back to dashboard” link. This is correct, but the empty state is sparse — a screenshot of what *would* be shown would be more useful.
Recommendation	Add a faded screenshot / illustration of a move card behind the empty-state heading, with a tooltip “Recommendations appear when our model spots a likely-positive move.”
Status	**Deferred.**

Individual Evaluation Templates

Each member’s individual heuristic evaluation template (Phase 1, before consolidation) is reproduced in the materials folder as `submission/materials/individual_heuristic_evaluations.md`.

Appendix G — Live Prototype + Repository

Artefact	URL
Live prototype	https://signal-drab-xi.vercel.app
Style guide route	https://signal-drab-xi.vercel.app/style-guide
User flow diagram	https://signal-drab-xi.vercel.app/flow
GitHub repository	https://github.com/Kazemkhani/HCI-Final-Project
Stage 5 commit	`d2f03c9` — fix(signal): wire every dead control — interactivity audit + fixes

DELIVERABLE 3 · PITCH DECK · 10 MARKS

The pitch.

The 12-slide pitch deck follows on the next pages, presented in landscape 16:9 format. It makes the case for Signal as we would pitch to a product manager or hackathon judge.